

Training Workers against a Frozen Synthesizer

Materials and splits. Each question q has a gold answer y^* and a pool of documents. Before any training, the pool is split once into two private document sets c_A and c_B , and the split is stored. Algorithms 1 and 2 and all evaluations use this same stored split. In our HotpotQA setting each question has ten documents: c_A contains one supporting document and four distractors, and c_B contains the other supporting document and four distractors. The workers are not told which document is supporting, and the order within each set is shuffled. Questions are divided into training, validation and test sets $\mathcal{D}_{\text{train}}$, $\mathcal{D}_{\text{val}}$ and $\mathcal{D}_{\text{test}}$. Validation and test questions are never used to train the workers or the synthesizer.

Agents. Both workers share one policy π_θ (LoRA parameters θ on a base model M) and differ only in their input $x_A = \text{PROMPT}(q, c_A)$ and $x_B = \text{PROMPT}(q, c_B)$. In this stage the synthesizer C is the frozen base model M ; it decodes greedily, so it acts as a deterministic part of the reward.

Evaluation protocol. $\text{EVALUATE}(\theta, C', \mathcal{D})$ (Algorithm 3) decodes both workers with the deterministic setting $\text{DECODE}_{\text{eval}}$, lets the synthesizer C' answer from the two reports, and returns the average F1 and exact match (EM) against y^* , together with the average number of tokens generated by A, B and C' . Reports are cached per worker checkpoint and data set, so different synthesizers are compared on identical reports, prompts and decoding settings. The variant $\text{EVALUATE}_\emptyset$ replaces both reports with empty strings. Checkpoints are selected by validation F1, and test results are computed once at the end.

Cross-paired reward. For each question, A and B each sample G reports from the policy used for data collection:

$$a_1, \dots, a_G \sim \pi_{\theta_{\text{old}}}(\cdot \mid x_A), \quad b_1, \dots, b_G \sim \pi_{\theta_{\text{old}}}(\cdot \mid x_B).$$

Every pair is answered by C and scored against the gold answer:

$$R_{ij} = \text{F1}(C(q, a_i, b_j), y^*), \quad i, j \in \{1, \dots, G\}. \quad (1)$$

Advantages. Each report is evaluated by its average reward over multiple partner reports, which reduces the dependence of its evaluation on any single partner:

$$Q_A(i) = \frac{1}{G} \sum_{j=1}^G R_{ij}, \quad Q_B(j) = \frac{1}{G} \sum_{i=1}^G R_{ij}. \quad (2)$$

Averaging does not remove this dependence: if all of B’s reports lack necessary information, a good report from A can still receive low rewards. A side carries a learning signal only if its marginal scores vary:

$$s_A = \mathbf{1}[\text{std}_k Q_A(k) > \delta], \quad s_B = \mathbf{1}[\text{std}_k Q_B(k) > \delta]. \quad (3)$$

For each side with signal, advantages are normalized within its group of G reports:

$$\text{Adv}_A(i) = \frac{Q_A(i) - \text{mean}_k Q_A(k)}{\text{std}_k Q_A(k) + \epsilon_n}, \quad \text{Adv}_B(j) = \frac{Q_B(j) - \text{mean}_k Q_B(k)}{\text{std}_k Q_B(k) + \epsilon_n}. \quad (4)$$

A question is used for training as soon as *at least one* side has signal ($s_A \vee s_B$); only the side(s) with signal contribute reports. A question with $s_A = s_B = 0$ contributes no policy gradient, so it

is discarded and another question is sampled in its place. This also covers the case where R varies but neither marginal does, e.g. $R = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$: success depends on the specific pairing, and no report is better on average.

Objective. For a report $o = (o_1, \dots, o_{|o|})$ generated from input x with advantage u , the token-level probability ratio and objective are

$$\rho_t(\theta) = \frac{\pi_\theta(o_t \mid x, o_{<t})}{\pi_{\theta_{\text{old}}}(o_t \mid x, o_{<t})}, \quad (5)$$

$$\ell_t(\theta; x, o, u) = \min\left(\rho_t(\theta) u, \text{clip}(\rho_t(\theta), 1 - \epsilon, 1 + \epsilon) u\right). \quad (6)$$

We use no KL penalty to a reference policy; the clipped ratio alone limits how far each update moves the policy. For each question q , the training set keeps only the reports of sides with signal, each with its input and advantage:

$$\mathcal{S}_q = \underbrace{\{(x_A, a_i, \text{Adv}_A(i))\}_{i=1}^G}_{\text{if } s_A=1} \cup \underbrace{\{(x_B, b_j, \text{Adv}_B(j))\}_{j=1}^G}_{\text{if } s_B=1}, \quad |\mathcal{S}_q| \in \{G, 2G\}. \quad (7)$$

Each element $(x, o, u) \in \mathcal{S}_q$ carries its own advantage: $u = \text{Adv}_A(i)$ for A’s reports and $u = \text{Adv}_B(j)$ for B’s reports. The objective averages over all reports in $\mathcal{S}_q$, from both sides, for every question in the update batch $\mathcal{Q}$ ($|\mathcal{Q}| = N$ questions with signal), and the loss is its negative:

$$\mathcal{J}(\theta) = \frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} \frac{1}{|\mathcal{S}_q|} \sum_{(x,o,u) \in \mathcal{S}_q} \frac{1}{|o|} \sum_{t=1}^{|o|} \ell_t(\theta; x, o, u), \quad (8)$$

$$\mathcal{L}(\theta) = -\mathcal{J}(\theta). \quad (9)$$

Only report tokens enter the loss: prompt and document tokens are conditioning only, and C receives no gradient. Tokens of A’s reports use $\text{Adv}_A(i)$, tokens of B’s reports use $\text{Adv}_B(j)$, and both update the shared parameters θ . The factor $1/|\mathcal{S}_q|$ averages over the reports retained for question q : when only one side has signal, the question contributes the mean over that side’s G reports, whereas keeping the other side’s zero-advantage reports in the average would halve this contribution. Because A and B share θ , an update computed from one side’s reports also changes the policy used by the other side.

Training the Synthesizer against Frozen Workers

Setup. The workers are frozen at the parameters θ^* selected on $\mathcal{D}_{\text{val}}$ by Algorithm 1. The synthesizer receives its own LoRA parameters ϕ on the same base model M ; we write C_ϕ for the trained synthesizer and $C_0 = M$ for the original one. Only ϕ is updated.

Data. For every training question, the frozen workers produce one report pair on the stored split, with the same decoding setting used at evaluation:

$$a = \text{DECODE}_{\text{eval}}(\pi_{\theta^*}, x_A), \quad b = \text{DECODE}_{\text{eval}}(\pi_{\theta^*}, x_B).$$

Each question contributes one example (q, a, b, y^*) to the dataset $\mathcal{D}_C$; the target is the gold answer only.

Algorithm 1 Cross-paired GRPO-style training of the workers against a frozen synthesizer

Input: $\mathcal{D}_{\text{train}}, \mathcal{D}_{\text{val}}$ with stored splits (q, y^*, c_A, c_B) ; base model M ; worker LoRA parameters θ ; group size G ; questions with signal per update N ; clip range ϵ ; threshold δ ; learning rate η ; steps T ; evaluation interval K

Output: worker parameters θ^* with the best validation F1

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1:  $C \leftarrow M$  ▷ frozen;  $C$  decodes greedily
2:  $\theta^* \leftarrow \theta; f^* \leftarrow -\infty$ 
3: for  $t = 1, \dots, T$  do
4:    $\pi_{\theta_{\text{old}}} \leftarrow \pi_{\theta}; \mathcal{S} \leftarrow \emptyset; n \leftarrow 0$  ▷  $n$ : questions with signal so far
5:   while  $n < N$  do ▷ collect  $N$  questions with signal
6:     Sample  $(q, y^*, c_A, c_B) \sim \mathcal{D}_{\text{train}}$  ▷ stored split
7:      $x_A \leftarrow \text{PROMPT}(q, c_A); x_B \leftarrow \text{PROMPT}(q, c_B)$ 
8:     Sample  $a_1, \dots, a_G \sim \pi_{\theta_{\text{old}}}(\cdot | x_A)$ 
9:     Sample  $b_1, \dots, b_G \sim \pi_{\theta_{\text{old}}}(\cdot | x_B)$ 
10:    for  $i = 1, \dots, G; j = 1, \dots, G$  do ▷  $G^2$  calls to  $C$ 
11:       $R_{ij} \leftarrow \text{F1}(C(q, a_i, b_j), y^*)$  ▷ Eq. (1)
12:    end for
13:     $Q_A(i) \leftarrow \frac{1}{G} \sum_j R_{ij}; Q_B(j) \leftarrow \frac{1}{G} \sum_i R_{ij}$  ▷ Eq. (2)
14:     $s_A \leftarrow \text{std}(Q_A) > \delta; s_B \leftarrow \text{std}(Q_B) > \delta$  ▷ Eq. (3)
15:    if  $s_A \vee s_B$  then ▷ one side with signal is enough
16:       $\mathcal{S}_q \leftarrow \emptyset$ 
17:      if  $s_A$  then
18:         $(\text{Adv}_A(1), \dots, \text{Adv}_A(G)) \leftarrow \text{NORM}(Q_A)$  ▷ Eq. (4)
19:         $\mathcal{S}_q \leftarrow \mathcal{S}_q \cup \{(x_A, a_i, \text{Adv}_A(i))\}_{i=1}^G$ 
20:      end if
21:      if  $s_B$  then
22:         $(\text{Adv}_B(1), \dots, \text{Adv}_B(G)) \leftarrow \text{NORM}(Q_B)$ 
23:         $\mathcal{S}_q \leftarrow \mathcal{S}_q \cup \{(x_B, b_j, \text{Adv}_B(j))\}_{j=1}^G$ 
24:      end if
25:       $\mathcal{S} \leftarrow \mathcal{S} \cup \{\mathcal{S}_q\}$  ▷ Eq. (7)
26:       $n \leftarrow n + 1$  ▷ counts once, one side or both
27:    else
28:      discard  $q$  ▷ no signal on either side
29:    end if
30:  end while
31:   $\theta \leftarrow \theta - \eta \nabla_{\theta} \mathcal{L}(\theta; \mathcal{S})$  ▷ Eqs. (8)–(9), report tokens only
32:  if  $t \bmod K = 0$  then ▷ checkpoint selection on validation
33:     $f \leftarrow \text{F1 returned by EVALUATE}(\theta, C, \mathcal{D}_{\text{val}})$ 
34:    if  $f > f^*$  then
35:       $f^* \leftarrow f; \theta^* \leftarrow \theta$ 
36:    end if
37:  end if
38: end for
39: return  $\theta^*$ 

40: function  $\text{NORM}(v \in \mathbb{R}^G)$ 
41:   return  $(v - \text{mean}(v)) / (\text{std}(v) + \epsilon_n)$ 
42: end function
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Objective. C_ϕ is trained with supervised fine-tuning. The loss is computed on the answer tokens only; the question and both reports are conditioning:

$$\mathcal{L}_{\text{SFT}}(\phi) = -\frac{1}{|\mathcal{D}_C|} \sum_{(q,a,b,y^*) \in \mathcal{D}_C} \frac{1}{|y^*|} \sum_{t=1}^{|y^*|} \log C_\phi(y_t^* \mid q, a, b, y_{<t}^*). \quad (10)$$

Evaluation. The checkpoint ϕ^* is selected by validation F1. On the test set, C_0 and C_{ϕ^*} are compared on identical cached reports (a_q, b_q) , prompts and decoding settings, with the reports and with both reports replaced by empty strings:

$$\text{F1}_{\text{rep}}(C') = \frac{1}{|\mathcal{D}_{\text{test}}|} \sum_{q \in \mathcal{D}_{\text{test}}} \text{F1}(C'(q, a_q, b_q), y_q^*), \quad (11)$$

$$\text{F1}_{\text{empty}}(C') = \frac{1}{|\mathcal{D}_{\text{test}}|} \sum_{q \in \mathcal{D}_{\text{test}}} \text{F1}(C'(q, \emptyset, \emptyset), y_q^*), \quad (12)$$

for $C' \in \{C_0, C_{\phi^*}\}$. EM is defined analogously, and we also report the average number of tokens generated by A, B and C' . A larger gain in F1_{rep} than in F1_{empty} is consistent with better use of the reports, but does not by itself establish this explanation: changes in output format, adaptation to the task, or the use of prior knowledge can also raise F1_{empty} .

Algorithm 2 Supervised fine-tuning of the synthesizer with frozen workers

Input: $\mathcal{D}_{\text{train}}, \mathcal{D}_{\text{val}}, \mathcal{D}_{\text{test}}$ with stored splits (q, y^*, c_A, c_B) , where y^* is the gold answer; frozen worker parameters θ^* from Algorithm 1; base model M ; synthesizer LoRA parameters ϕ ; learning rate η ; epochs E

Output: synthesizer parameters ϕ^* with the best validation F1; test results of C_0 and C_{ϕ^*} with and without reports

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1:  $C_0 \leftarrow M$  ▷ original synthesizer, kept for comparison
   Build the SFT data
2:  $\mathcal{D}_C \leftarrow \emptyset$ 
3: for each  $(q, y^*, c_A, c_B) \in \mathcal{D}_{\text{train}}$  do ▷ stored split
4:    $a \leftarrow \text{DECODE}_{\text{eval}}(\pi_{\theta^*}, \text{PROMPT}(q, c_A))$  ▷ same setting as evaluation
5:    $b \leftarrow \text{DECODE}_{\text{eval}}(\pi_{\theta^*}, \text{PROMPT}(q, c_B))$ 
6:    $\mathcal{D}_C \leftarrow \mathcal{D}_C \cup \{(q, a, b, y^*)\}$  ▷ target: gold answer  $y^*$ 
7: end for
   Train only the synthesizer, select on validation
8:  $\phi^* \leftarrow \phi$ ;  $f^* \leftarrow -\infty$ 
9: for  $e = 1, \dots, E$  do
10:  for each minibatch  $\mathcal{U} \subset \mathcal{D}_C$  do
11:     $\phi \leftarrow \phi - \eta \nabla_{\phi} \mathcal{L}_{\text{SFT}}(\phi; \mathcal{U})$  ▷ Eq. (10), answer tokens only
12:  end for
13:   $f \leftarrow \text{F1}$  returned by  $\text{EVALUATE}(\theta^*, C_{\phi}, \mathcal{D}_{\text{val}})$ 
14:  if  $f > f^*$  then
15:     $f^* \leftarrow f$ ;  $\phi^* \leftarrow \phi$ 
16:  end if
17: end for
   Final comparison on held-out test questions
18: for  $C' \in \{C_0, C_{\phi^*}\}$  do ▷ identical cached reports
19:    $\text{EVALUATE}(\theta^*, C', \mathcal{D}_{\text{test}})$ ;  $\text{EVALUATE}_{\emptyset}(\theta^*, C', \mathcal{D}_{\text{test}})$  ▷ Eqs. (11)–(12)
20: end for
21: return  $\phi^*$  and the four test results (F1, EM, tokens)
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Algorithm 3 Evaluation procedure shared by Algorithms 1 and 2

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1: function  $\text{EVALUATE}(\theta, C', \mathcal{D})$ 
2:   for each  $(q, y^*, c_A, c_B) \in \mathcal{D}$  do ▷ stored split
3:      $a \leftarrow \text{DECODE}_{\text{eval}}(\pi_{\theta}, \text{PROMPT}(q, c_A))$  ▷ reports cached per  $(\theta, \mathcal{D})$ 
4:      $b \leftarrow \text{DECODE}_{\text{eval}}(\pi_{\theta}, \text{PROMPT}(q, c_B))$ 
5:      $\hat{y} \leftarrow C'(q, a, b)$  ▷  $C'(q, \emptyset, \emptyset)$  in  $\text{EVALUATE}_{\emptyset}$ 
6:     Record  $\text{F1}(\hat{y}, y^*)$ ,  $\text{EM}(\hat{y}, y^*)$ ,  $|a|$ ,  $|b|$ ,  $|\hat{y}|$ 
7:   end for
8:   return averages: F1, EM, tokens generated by A, B and  $C'$ 
9: end function
```
